claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ad03fb07a6b3a8d58.jsonl`
- SHA-256: `acc91081fc51f65e6b2680f2f48d2929050a3391e314a27ed01d167523d4021c`
- Source modified: `2026-06-06T17:31:31+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:37.973Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/html/2511.17045v2
Title: RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis (AAAI 2026)
Found via: recent/applied search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:40.876Z)

I'll fetch the source and extract relevant claims.

3. user (2026-06-06T17:30:57.918Z)

RacketVision: Multi-Sport Racket Analysis Benchmark

1. Paper Overview & Benchmark

RacketVision introduces a new dataset for sports analytics. The authors state: "We introduce RacketVision, a novel dataset and benchmark for advancing computer vision in sports analytics, covering table tennis, tennis, and badminton."

The benchmark addresses gaps in existing research: "existing datasets focus narrowly on ball tracking within a single sport at a time, falling short in two critical aspects: 1) They fail to leverage shared ball motion patterns across different sports. 2) Despite the racket being a central component, racket-specific annotations and analysis are lacking."

2. Sports & Tasks Covered

The paper covers three sports with multiple interconnected tasks: "RacketVision comprises a collection of 1,672 video clips (435,179 frames, 12,755 seconds) spanning badminton, tennis, and table tennis."

Tasks include: "three interconnected tasks with corresponding annotations: fine-grained ball tracking, articulated racket pose estimation, and predictive ball trajectory forecasting."

3. Dataset Specifications & Licenses

The paper provides no explicit license information for the released dataset. Dataset composition includes: "50 games of table tennis (780 clips, 170,027 frames), 431 tennis games (431 clips, 150,399 frames), and 461 badminton games (461 clips, 114,753 frames)."

Data sourcing: "video from 942 top-level professional game broadcasts on YouTube" with "20% of frames within each clip evenly sampled for manual labeling."

4. Performance Results

Ball Tracking (best multi-sport model MS-TrackNetV3):
- Table Tennis: Precision 0.924, Recall 0.762, MDE 3.41px, mAP 71.1
- Tennis: Precision 0.945, Recall 0.880, MDE 1.96px, mAP 81.9
- Badminton: Precision 0.915, Recall 0.865, MDE 1.54px, mAP 83.1

Racket Pose Estimation (MS RTMPose):
- Table Tennis: PCK@0.2 81.8%, MPJPE 9.71px
- Tennis: PCK@0.2 89.6%, MPJPE 5.34px
- Badminton: PCK@0.2 88.5%, MPJPE 5.00px

Trajectory Prediction (LSTM with Cross-Attention):
Short predictions (N=20, M=5) show improvements; long predictions (N=80, M=20) demonstrate effectiveness of Cross-Attention fusion over naive concatenation.

5. Compute Requirements

The paper does not specify GPU or computational requirements for training or inference.

6. Publication Date

Published November 27, 2025 (generated timestamp indicates arxiv version 2511.17045v2).

7. Authors & Affiliation

"Linfeng Dong, Yuchen Yang, Hao Wu, Wei Wang, Yuenan Hou, Zhihang Zhong, Xiao Sun" with affiliations at Shanghai AI Laboratory. Work completed during internships at this institution.

4. assistant (2026-06-06T17:31:01.373Z)

Let me verify a couple of key details directly from the source to ensure quote accuracy on the segmentation relevance and the TrackNet baseline.

5. user (2026-06-06T17:31:09.646Z)

Analysis of RacketVision Paper

(a) Rally/Point Temporal Segmentation

The paper does **not** address rally or point segmentation. The scope is limited to perception and prediction tasks. Relevant quote:

"A clip is defined as a continuous segment of 5-10 seconds where the ball is actively in play, which focuses the dataset on the most analytically relevant portions of the game."

The paper assumes clips are pre-segmented; it does not detect when rallies begin or end.

(b) TrackNetV3 Baseline

The paper uses TrackNetV3 as a ball tracking baseline. Key description:

"TrackNetV3 (chen2023tracknetv3): A specialized heatmap-based network for tracking small, high-speed objects in sports, which can leverage temporal context from multiple frames."

(c) Multi-Sport vs Single-Sport Training

Multi-sport training **does improve** ball tracking generalization:

"By comparing the best multi-sport model (MS-TrackNetV3, #F=4, bold results) with the best single-sport model (TrackNetV3, #F=4, underlined results), we observe a clear trend of improved generalization. For example, the MS model boosts mAP by a significant 19.2% in Tennis (81.9 vs. 68.7) and 14.6% in Badminton (83.1 vs. 72.5)."

(d) Audio Mention

No mention of audio appears in the provided document. The paper focuses exclusively on visual analysis.

6. assistant (2026-06-06T17:31:12.823Z)

I have enough to extract claims. The source is primary research (AAAI 2026 paper from Shanghai AI Laboratory). It's relevant to the substrate (ball tracking) but notably does NOT address the core research question of temporal segmentation.

7. user (2026-06-06T17:31:25.007Z)

Structured output provided successfully

8. assistant (2026-06-06T17:31:31.548Z)

Extracted 5 claims from the RacketVision paper (arXiv:2511.17045v2, AAAI 2026, Shanghai AI Laboratory).

Key finding for the research question: this is a primary research source, but it does NOT address rally/point temporal segmentation ? the paper's central scope is perception (ball tracking, racket pose, trajectory forecasting) on clips that are already pre-segmented into 5-10s active-play windows. So it is relevant to our WASB-style shuttle-trajectory substrate (and offers a recent badminton ball-tracking recall reference of 0.865 plus evidence that multi-sport joint training boosts badminton mAP by 14.6%), and it introduces racket-pose estimation as a potential complementary local feature ? but it provides no method for the start/end segmentation that is our core problem. It also carries license/provenance risk (YouTube-sourced, no explicit license stated).